Vandermond Matrix:

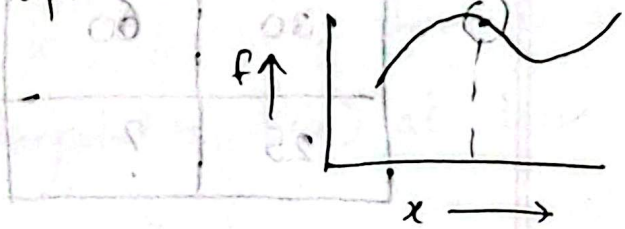
$$P_n(x) = \sum_{k=0}^{n} a_k x^k$$

$$P_2(x) = a_0 + a_1 x + a_2 x^2$$

$$= 5 + 6x + 7x^2$$

$$P_2(x=2) = 5 + 6 \cdot 2 + 7 \cdot (2)^2 = \square \quad \text{[single node value]}$$

Vandermond matrix talks about multiple values.



$$P_n(x_0) = a_0 + a_1 x_0 + a_2 x_0^2 + a_3 x_0^3 + \ldots + a_n x_0^n$$

$$\vdots$$

$$P_n(x_n) = a_0 + a_1 x_n + a_2 x_n^2 + a_3 x_n^3 + \ldots + a_n x_n^n$$

$V \cdot a = f$ → function matrix; coefficient matrix

$$\begin{bmatrix} 1 & x_0 & x_0^2 & x_0^3 & \ldots & x_n^n \\ 1 & x_1 & x_1^2 & x_1^3 & \ldots & x_1^n \\ \vdots & & & & & \\ 1 & x_n & x_n^2 & x_n^3 & \ldots & x_n^n \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} P_n(x_0) \\ P_n(x_1) \\ \vdots \\ P_n(x_n) \end{bmatrix}$$

Vandermonde Matrix ~~x_arr~~

Coefficient/ Taylor coefficient

f
p_x_arr
data_y

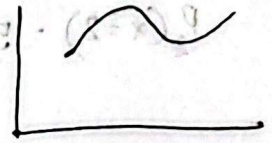
Matrix V must be invertible [$\det(V) \neq 0$]

$$\det(A) = \prod_{0 \le i < j \le n} (x_i - x_j)$$

$a = V^{-1} f$ ✓

$a = f V^{-1}$ ✗

age (x)	weight (y)
10	40
30	60
25	?



$P_1(x) = a_0 + a_1 x$

$x_0 \longrightarrow a_0 + a_1 10 = 40$

$x_1 \longrightarrow a_0 + a_1 30 = 60$

$$Va = f$$

$$\Rightarrow \begin{bmatrix} 1 & 10 \\ 1 & 30 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 40 \\ 60 \end{bmatrix}$$

$$\Rightarrow a = V^{-1} f$$

$$= \begin{bmatrix} 1 & 10 \\ 1 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 40 \\ 60 \end{bmatrix}$$

$$= \begin{bmatrix} 30 \\ 1 \end{bmatrix}$$

$a_0 = 30, \quad a_1 = 1$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2$$

Compare with Taylor

$P_1(x) = 30 + 1x$

Ans: $c_0 = 30$

$c_1 = 1$

$3x^2 + 2x + 1$

$c_0 = 1$

$c_1 = 2$

$c_2 = 3$

Reason

$f'(x) = 6x + 2$

$f'(0) = 2$

$f''(x) = 6/2!$

$f''(0) = 6/2!$

$= 3$

* Same value নিবো না কেন

$P_1(x) = 30 + x_1$

$= 30 + 25$

$= 55$ (Ans.)

Theorem:

- **Existence / Uniqueness:**

Given $(n+1)$ distinct nodes $x_0, x_1, \ldots, x_n$ there is a unique polynomial $P_n \in P_n$ that interpolates $f(x)$ at these nodes

Time	velocity
15	362.8
20	517.35

$P_1(x) = a_0 + a_1 x$

$P_1(15) = a_0 + 15a_1 = 362.8$

$P_1(20) = a_0 + 20a_1 = 517.35$

Now,

$$\begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1 & 15 \\ 1 & 20 \end{bmatrix}^{-1} \begin{bmatrix} 362.8 \\ 517.35 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} -100.85 \\ 30.91 \end{bmatrix}$$

$\therefore P_1(x) = -100.85 + 30.91x$

Acceleration চাইলে differentiate

$f(x) = \cos x$ nodes $\{0, \pi/2, \pi\}$

$x_0 = 0 \quad x_1 = \frac{\pi}{2} \quad x_2 = \pi$

$P_2(x) = a_0 + a_1 x + a_2 x^2$

Ans:

$f(x_0) = 1 \quad , \quad f(x_1) = 0 \quad , \quad f(x_2) = -1$

3 nodes $\longrightarrow P_2(x)$

$\downarrow$

(3×3) Vandermonde matrix

Now,

$$\begin{bmatrix} 1 & 0 & 0^2 \\ 1 & \pi/2 & (\pi/2)^2 \\ 1 & \pi & \pi^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0^2 \\ 1 & \pi/2 & (\pi/2)^2 \\ 1 & \pi & \pi^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2/\pi \\ 0 \end{bmatrix}$$

So,

$$P_2(x) = a_0 + a_1 x + a_2 x^2$$

$$= 1 + \left(\frac{-2}{\pi}\right) x + 0x^2$$

$$= 1 - \frac{2}{\pi} x$$